

$$= \sum_{n=-\infty}^{\infty} (-1)^n x^{n(3n+1)/2},$$

which was to be proved.

We make a final comment about the partition function $p(n)$. The nature of its growth as $n \rightarrow \infty$ can be analyzed in terms of the behavior of $1/\prod_{n=1}^{\infty} (1-x)^n$ as $|x| \rightarrow 1$. In fact, by elementary considerations, we can get a rough order of growth of $p(n)$ from the growth of the generating function as $x \rightarrow 1$; see Exercises 5 and 6. A more refined analysis requires the transformation properties of the generating function which goes back to the corresponding Proposition 1.9 about η . This leads to a very good asymptotic formula for $p(n)$. It may be found in Appendix A.

3 The theorems about sums of squares

The ancient Greeks were fascinated by triples of integers (a, b, c) that occurred as sides of right triangles. These are the “Pythagorean triples,” which satisfy $a^2 + b^2 = c^2$. According to Diophantus of Alexandria (ca. 250 AD), if c is an integer of the above kind, and a and b have no common factors (a case to which one may easily reduce), then c is the sum of two squares, that is, $c = m^2 + n^2$ with $m, n \in \mathbb{Z}$; and conversely, any such c arises as the hypotenuse of a triangle whose sides are given by a Pythagorean triple (a, b, c) . (See Exercise 8.) Therefore, it is natural to ask the following question: which integers can be written as the sum of two squares? It is easy to see that no number of the form $4k + 3$ can be so written, but to determine which integers can be expressed in this way is not obvious.

Let us pose the question in a more quantitative form. We define $r_2(n)$ to be the number of ways n can be written as the sum of two squares, counting obvious repetitions; that is, $r_2(n)$ is the number of pairs (x, y) , $x, y \in \mathbb{Z}$, so that

$$n = x^2 + y^2.$$

For example, $r_2(3) = 0$, but $r_2(5) = 8$ because $5 = (\pm 2)^2 + (\pm 1)^2$, and also $5 = (\pm 1)^2 + (\pm 2)^2$. Hence, our first problem can be posed as follows:

Sum of two squares: Which integers can be written as a sum of two squares? More precisely, can one determine an expression for $r_2(n)$?

Next, since not every positive integer can be expressed as the sum of two squares, we may ask if three squares, or possibly four squares suffice.

However, the fact is that there are infinitely many integers that cannot be written as the sum of three squares, since it is easy to check that no integer of the form $8k + 7$ can be so written. So we turn to the question of four squares and define, in analogy with $r_2(n)$, the function $r_4(n)$ to be the number of ways of expressing n as a sum of four squares. Therefore, a second problem that arises is:

Sum of four squares: Can every positive integer be written as a sum of four squares? More precisely, determine a formula for $r_4(n)$.

It turns out that the problems of two squares and four squares, which go back to the third century, were not resolved until about 1500 years later, and their full solution was first given by the use of Jacobi's theory of theta functions!

3.1 The two-squares theorem

The problem of representing an integer as the sum of two squares, while obviously additive in nature, has a nice multiplicative aspect: if n and m are two integers that can be written as the sum of two squares, then so can their product nm . Indeed, suppose $n = a^2 + b^2$, $m = c^2 + d^2$, and consider the complex number

$$x + iy = (a + ib)(c + id).$$

Clearly, x and y are integers since $a, b, c, d \in \mathbb{Z}$, and by taking absolute values on both sides we see that

$$x^2 + y^2 = (a^2 + b^2)(c^2 + d^2),$$

and it follows that $nm = x^2 + y^2$.

For these reasons the divisibility properties of n play a crucial role in determining $r_2(n)$. To state the basic result we define two new **divisor functions**: we let $d_1(n)$ denote the number of divisors of n of the form $4k + 1$, and $d_3(n)$ the number of divisors of n of the form $4k + 3$. The main result of this section provides a complete answer to the two-squares problem:

Theorem 3.1 *If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.*

A direct consequence of the above formula for $r_2(n)$ may be stated as follows. If $n = p_1^{a_1} \cdots p_r^{a_r}$ is the prime factorization of n where $p_1, \dots, p_r$ are distinct, then:

The positive integer n can be represented as the sum of two squares if and only if every prime p_j of the form $4k + 3$ that occurs in the factorization of n has an even exponent a_j .

The proof of this deduction is outlined in Exercise 9.

To prove the theorem, we first establish a crucial relationship that identifies the generating function of the sequence $\{r_2(n)\}_{n=1}^{\infty}$ with the square of the θ function, namely

$$(6) \quad \theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n,$$

whenever $q = e^{\pi i \tau}$ with $\tau \in \mathbb{H}$. The proof of this identity relies simply on the definition of r_2 and θ . Indeed, if we first recall that $\theta(\tau) = \sum_{n=-\infty}^{\infty} q^{n^2}$, then we obtain

$$\begin{aligned} \theta(\tau)^2 &= \left(\sum_{n_1=-\infty}^{\infty} q^{n_1^2} \right) \left(\sum_{n_2=-\infty}^{\infty} q^{n_2^2} \right) \\ &= \sum_{(n_1, n_2) \in \mathbb{Z} \times \mathbb{Z}} q^{n_1^2 + n_2^2} \\ &= \sum_{n=0}^{\infty} r_2(n) q^n, \end{aligned}$$

since $r_2(n)$ counts the number of pairs (n_1, n_2) with $n_1^2 + n_2^2 = n$.

Proposition 3.2 *The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities*

$$(7) \quad \theta(\tau)^2 = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}},$$

whenever $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$.

Proof. We note first that both series converge absolutely since $|q| < 1$, and the first equals the second, because $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$.

Since $(1 + q^{2n})^{-1} = (1 - q^{2n})/(1 - q^{4n})$, the right-hand side of (7) equals

$$1 + 4 \sum_{n=1}^{\infty} \left(\frac{q^n}{1 - q^{4n}} - \frac{q^{3n}}{1 - q^{4n}} \right).$$

However, since $1/(1 - q^{4n}) = \sum_{m=0}^{\infty} q^{4nm}$, we have

$$\sum_{n=1}^{\infty} \frac{q^n}{1 - q^{4n}} = \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k) q^k,$$

because $d_1(k)$ counts the number of divisors of k that are of the form $4m + 1$. Observe that the series $\sum d_1(k) q^k$ converges since $d_1(k) \leq k$.

A similar argument shows that

$$\sum_{n=1}^{\infty} \frac{q^{3n}}{1 - q^{4n}} = \sum_{k=1}^{\infty} d_3(k) q^k,$$

and the proof of the proposition is complete.

In effect, we see that the identity (6) links the original problem in arithmetic with the problem in complex analysis of establishing the relation (7).

We shall now find it convenient to use $\mathcal{C}(\tau)$ to denote³

$$(8) \quad \mathcal{C}(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(n\pi\tau)},$$

where $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$. Our work then becomes to prove the identity $\theta(\tau)^2 = \mathcal{C}(\tau)$.

What is truly remarkable are the different yet parallel ways that the functions θ and $\mathcal{C}$ arise. The genesis of the function θ may be thought to be the heat diffusion equation on the real line; the corresponding heat kernel is given in terms of the Gaussian $e^{-\pi x^2}$ which is its own Fourier transform; and finally the transformation rule for θ results from the Poisson summation formula.

The parallel with $\mathcal{C}$ is that it arises from another differential equation: the steady-state heat equation in a strip; there, the corresponding kernel is $1/\cosh \pi x$ (Section 1.3, Chapter 8), which again is its own Fourier transform (Example 3, Chapter 3). The transformation rule for $\mathcal{C}$ results, once again, from the Poisson summation formula.

To prove the identity $\theta^2 = \mathcal{C}$ we will first show that these two functions satisfy the same structural properties. For θ^2 we had the transformation law $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7).

³We denote the function by $\mathcal{C}$ because we are summing a series of cosines.

An identical transformation law holds for $\mathcal{C}(\tau)$! Indeed, if we set $a = 0$ in the relation (5) of Chapter 4 we obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi n t)} = \frac{1}{t} \sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi n/t)}.$$

This is precisely the identity

$$\mathcal{C}(\tau) = (i/\tau) \mathcal{C}(-1/\tau)$$

for $\tau = it$, $t > 0$, which therefore also holds for all $\tau \in \mathbb{H}$ by analytic continuation.

It is also obvious from their definitions that both $\theta(\tau)^2$ and $\mathcal{C}(\tau)$ tend to 1 as $\text{Im}(\tau) \rightarrow \infty$. The last property we want to examine is the behavior of both functions at the “cusp” $\tau = 1$.⁴

For θ^2 we shall invoke Corollary 1.8 to see that $\theta(1 - 1/\tau)^2 \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\text{Im}(\tau) \rightarrow \infty$.

For $\mathcal{C}$ we can do the same, again using the Poisson summation formula. In fact, if we set $a = 1/2$ in equation (5), Chapter 4, we find

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{\cosh(\pi n/t)} = t \sum_{n=-\infty}^{\infty} \frac{1}{\cosh(\pi(n + 1/2)t)}.$$

Therefore, by analytic continuation we deduce that

$$\mathcal{C}(1 - 1/\tau) = \left(\frac{\tau}{i}\right) \sum_{n=-\infty}^{\infty} \frac{1}{\cos(\pi(n + 1/2)\tau)}.$$

The main terms of this sum are those for $n = -1$ and $n = 0$. This easily gives

$$\mathcal{C}(1 - 1/\tau) = 4 \left(\frac{\tau}{i}\right) e^{\pi i\tau/2} + O(|\tau|e^{-3\pi t/2}), \quad \text{as } t \rightarrow \infty,$$

and where $\tau = \sigma + it$. We summarize our conclusions in a proposition.

Proposition 3.3 *The function $\mathcal{C}(\tau) = \sum 1/\cos(\pi n\tau)$, defined in the upper half-plane, satisfies*

(i) $\mathcal{C}(\tau + 2) = \mathcal{C}(\tau)$.

(ii) $\mathcal{C}(\tau) = (i/\tau)\mathcal{C}(-1/\tau)$.

⁴Why we refer to the point $\tau = 1$ as a cusp, and the reason for its importance, will become clear later on.

(iii) $\mathcal{C}(\tau) \rightarrow 1$ as $\text{Im}(\tau) \rightarrow \infty$.

(iv) $\mathcal{C}(1 - 1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\text{Im}(\tau) \rightarrow \infty$.

Moreover, $\theta(\tau)^2$ satisfies the same properties.

With this proposition, we prove the identity of $\theta(\tau)^2 = \mathcal{C}(\tau)$ with the aid of the following theorem, in which we shall ultimately set $f = \mathcal{C}/\theta^2$.

Theorem 3.4 *Suppose f is a holomorphic function in the upper half-plane that satisfies:*

(i) $f(\tau + 2) = f(\tau)$,

(ii) $f(-1/\tau) = f(\tau)$,

(iii) $f(\tau)$ is bounded,

then f is constant.

For the proof of this theorem, we introduce the following subset of the closed upper half-plane, which is defined by

$$\mathcal{F} = \{\tau \in \overline{\mathbb{H}} : |\text{Re}(\tau)| \leq 1 \text{ and } |\tau| \geq 1\},$$

and illustrated in Figure 1.

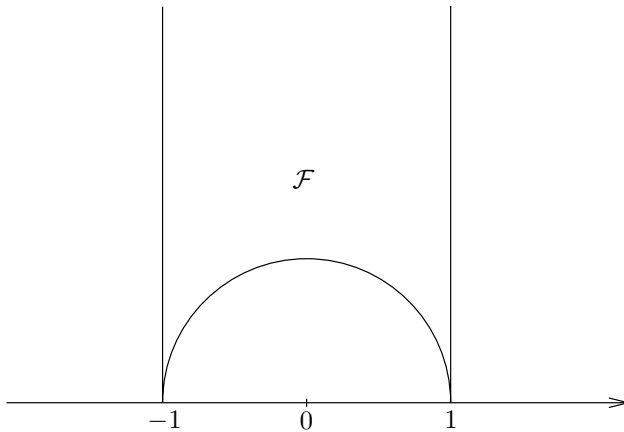


Figure 1. The domain $\mathcal{F}$

The points corresponding to $\tau = \pm 1$ are called **cusps**. They are equivalent under the mapping $\tau \mapsto \tau + 2$.

Lemma 3.5 *Every point in the upper half-plane can be mapped into $\mathcal{F}$ using repeatedly one or another of the following fractional linear transformations or their inverses:*

$$T_2 : \tau \mapsto \tau + 2, \quad S : \tau \mapsto -1/\tau.$$

For this reason, $\mathcal{F}$ is called the **fundamental domain**⁵ for the group of transformations generated by T_2 and S .

In fact, we let G denote the group generated by T_2 and S . Since T_2 and S are fractional linear transformations, we may represent an element $g \in G$ by a matrix

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

with the understanding that

$$g(\tau) = \frac{a\tau + b}{c\tau + d}.$$

Since the matrices representing T_2 and S have integer coefficients and determinant 1, the same is true for all matrices of elements in G . In particular, if $\tau \in \mathbb{H}$, then

$$(9) \quad \operatorname{Im}(g(\tau)) = \frac{\operatorname{Im}(\tau)}{|c\tau + d|^2}.$$

Proof of Lemma 3.5. Let $\tau \in \mathbb{H}$. If $g \in G$ with $g(\tau) = (a\tau + b)/(c\tau + d)$, then c and d are integers, and by (9) we may choose a $g_0 \in G$ such that $\operatorname{Im}(g_0(\tau))$ is maximal. Since the translations T_2 and their inverses do not change imaginary parts, we may apply finitely many of them to see that there exists $g_1 \in G$ with $|\operatorname{Re}(g_1(\tau))| \leq 1$ and $\operatorname{Im}(g_1(\tau))$ is maximal. It now suffices to prove that $|g_1(\tau)| \geq 1$ to conclude that $g_1(\tau) \in \mathcal{F}$. If this were not true, that is, $|g_1(\tau)| < 1$, then $\operatorname{Im}(Sg_1(\tau))$ would be greater than $\operatorname{Im}(g_1(\tau))$ since

$$\operatorname{Im}(Sg_1(\tau)) = \operatorname{Im}(-1/g_1(\tau)) = -\frac{\operatorname{Im}(\overline{g_1(\tau)})}{|g_1(\tau)|^2} > \operatorname{Im}(g_1(\tau)),$$

and this contradicts the maximality of $\operatorname{Im}(g_1(\tau))$.

We can now prove the theorem. Suppose f is not constant, and let $g(z) = f(\tau)$ where $z = e^{\pi i \tau}$. The function g is well defined for z in the

⁵Strictly speaking, the notion of a fundamental domain requires that every point have a unique representative in the domain. In the present case, ambiguity arises only for points that are on the boundary of $\mathcal{F}$.

punctured unit disc, since f is periodic of period 2, and moreover, g is bounded near the origin by assumption (iii) of the theorem. Hence 0 is a removable singularity for g , and $\lim_{z \rightarrow 0} g(z) = \lim_{\text{Im}(\tau) \rightarrow \infty} f(\tau)$ exists. So by the maximum modulus principle,

$$\lim_{\text{Im}(\tau) \rightarrow \infty} |f(\tau)| < \sup_{\tau \in \mathcal{F}} |f(\tau)|.$$

Now we must investigate the behavior of f at the points $\tau = \pm 1$. Since $f(\tau + 2) = f(\tau)$, it suffices to consider the point $\tau = 1$. We claim that

$$\lim_{\text{Im}(\tau) \rightarrow \infty} f(1 - 1/\tau)$$

exists and moreover

$$\lim_{\text{Im}(\tau) \rightarrow \infty} |f(1 - 1/\tau)| < \sup_{\tau \in \mathcal{F}} |f(\tau)|.$$

The argument is essentially the same as the one above, except that we first need to interchange $\tau = 1$ with the point at infinity. In other words, we wish to investigate the behavior of $F(\tau) = f(1 - 1/\tau)$ for τ near ∞ . The important step is to prove that F is periodic. To this end, we consider the fractional linear transformation associated to the matrix

$$U_n = \begin{pmatrix} 1 - n & n \\ -n & 1 + n \end{pmatrix},$$

that is,

$$\tau \mapsto \frac{(1 - n)\tau + n}{-n\tau + (1 + n)},$$

which maps 1 to 1. Now let $\mu(\tau) = 1/(1 - \tau)$ which maps 1 to ∞ , and whose inverse $\mu^{-1}(\tau) = 1 - 1/\tau$ takes ∞ to 1. Then

$$U_n = \mu^{-1} T_n \mu,$$

where T_n is the translation $T_n(\tau) = \tau + n$. As a consequence,

$$U_n U_m = U_{n+m},$$

and

$$U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = T_2 S.$$

Thus any U_n can be obtained by finitely many applications of T_2 , S , or their inverses. Since f is invariant under T_2 and S , it is also invariant under U_m . So we find that

$$f(\mu^{-1}T_n\mu(\tau)) = f(\tau).$$

Therefore, if we let $F(\tau) = f(\mu^{-1}(\tau)) = f(1 - 1/\tau)$, we find that F is periodic of period 1, that is,

$$F(T_n\tau) = F(\tau) \quad \text{for every integer } n.$$

Now, by the previous argument, if we set $h(z) = F(\tau)$ with $z = e^{2\pi i\tau}$, we see that h has a removable singularity at $z = 0$, and the desired inequality follows by the maximum principle.

We conclude from this analysis that f attains its maximum in the interior of the upper half-plane, and this contradicts the maximum principle.

The proof of the two-squares theorem is now only one step away.

We consider the function $f(\tau) = \mathcal{C}(\tau)/\theta(\tau)^2$. Since we know by the product formula that $\theta(\tau)$ does not vanish in the upper half-plane (Corollary 1.4), we find that f is holomorphic in $\mathbb{H}$. Moreover, by Proposition 3.3, f is invariant under the transformations T_2 and S , that is, $f(\tau + 2) = f(\tau)$ and $f(-1/\tau) = f(\tau)$. Finally, in the fundamental domain $\mathcal{F}$, the function $f(\tau)$ is bounded, and in fact tends to 1 as $\text{Im}(\tau)$ tends to infinity, or as τ tends to the cusps ± 1 . This is because of properties (iii) and (iv) in Proposition 3.3, which are verified by both $\mathcal{C}$ and θ^2 . Thus f is bounded in $\mathbb{H}$. The result is that f is a constant, which must be 1, proving that $\theta(\tau)^2 = \mathcal{C}(\tau)$, and with it the two-squares theorem.

3.2 The four-squares theorem

Statement of the theorem

In the rest of this chapter, we shall consider the case of four squares. More precisely, we will prove that every positive integer is the sum of four squares, and moreover we will determine a formula for $r_4(n)$ that describes the number of ways this can be done.

We need to introduce another divisor function, which we denote by $\sigma_1^*(n)$, and which equals the sum of divisors of n that are not divisible by 4. The main theorem we shall prove is the following.

Theorem 3.6 *Every positive integer is the sum of four squares, and moreover*

$$r_4(n) = 8\sigma_1^*(n) \quad \text{for all } n \geq 1.$$